



Lesson 4. Continuous Variables

- 1. Continuous Variable Characterization**
- 2. Mathematical expectation**
- 3. Others Parameters of Position**
- 4. Variance**
- 5. Main distributions of probability**
- 6. Uniform Distribution**
- 7. Exponential Distribution**
- 8. Normal Distribution**
- 9. X^2 -Distribution**
- 10. t- Distribution**
- 11. F-Distribution**
- 12. Approaches between discrete distributions**

Continuous Variable Characterization

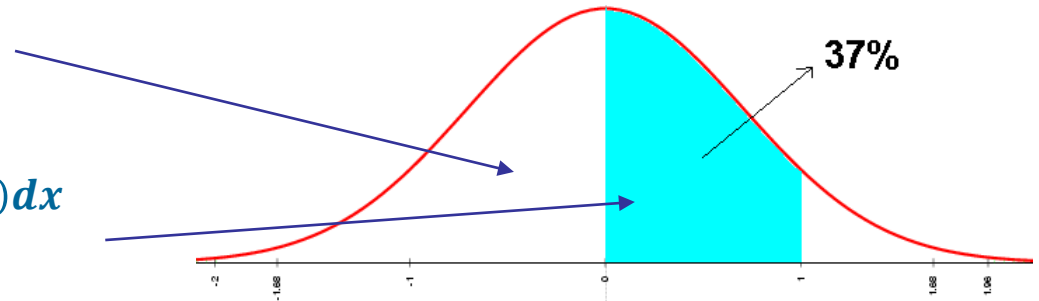
How to characterize the probability distribution of a continuous variable X is based on its density function ($f(x)$) whose fundamental property is that **the area under the density function of a random variable between two values 'a' and 'b', coincides with the probability that the random variable has values in that range (the probability of a particular value is zero but it doesn't have to be that of a finite or infinite interval).**

1. Infinite

$$\forall x \in \mathbb{R} \quad P(X \leq x) = \int_{-\infty}^x f(x)dx$$

2. Finite

$$\forall [a, b] \in \mathbb{R} \quad P(X \in [a, b]) = \int_a^b f(x)dx$$

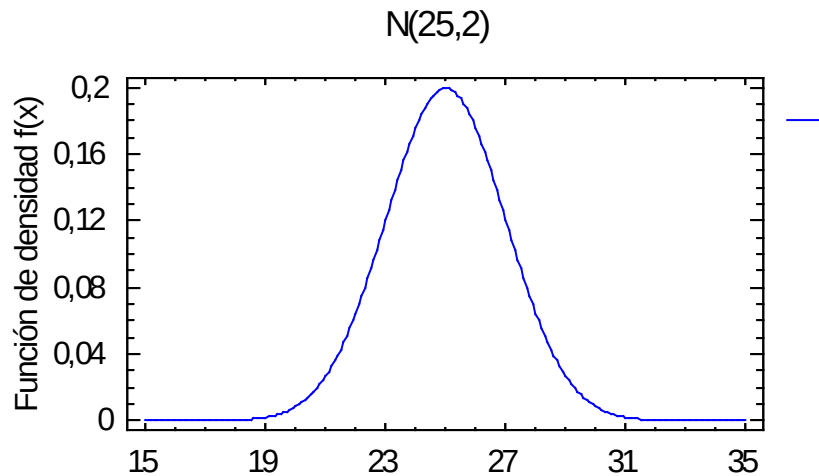


(*) The density function $f(x)$ is non-zero only in the domain or sample space of the variable, is not negative and the integral between - and + infinity is 1 (Probability of the sample space of the variable)

Continuous Variable Characterization

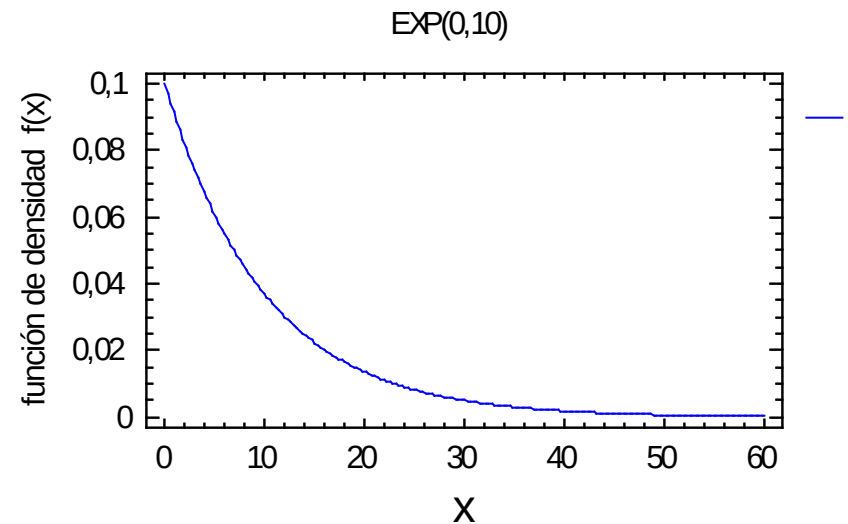
Normal $X = N(\mu, \sigma)$

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \quad -\infty < x < +\infty$$



Exponential $X = \text{Exp}(\theta)$

$$f(x) = \theta e^{-\theta x} \quad \forall x \geq 0$$

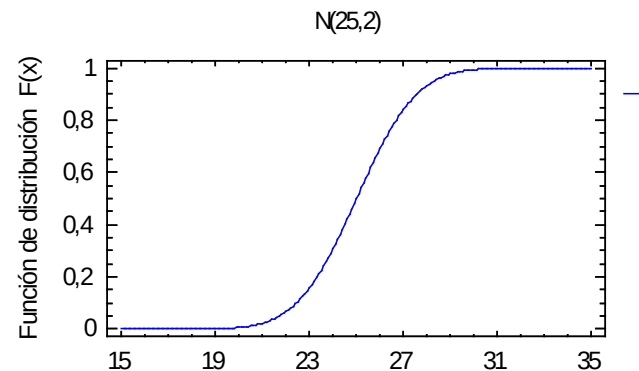


Continuous Variable Characterisation

By means of the Distribution Function that gives the accumulated probability up to each possible value of the variable.

$$F(x) = P(X \leq x) \quad \forall x \in]-\infty, +\infty[$$

$$F(x) = \int_{-\infty}^x f(x)dx \quad \forall x \in]-\infty, +\infty[$$



Relation between F(x) and f(x)

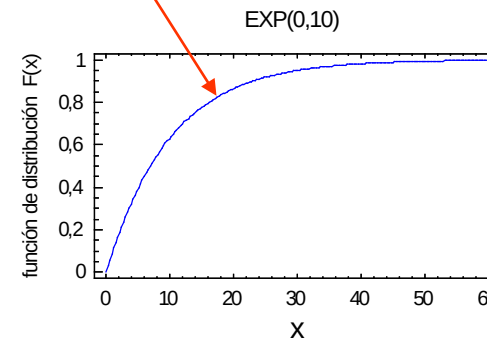
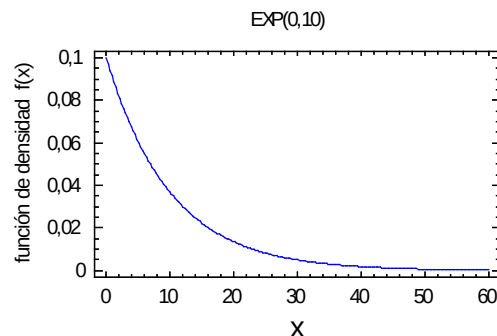
$$F(x) = P(X \leq x) = \int_{-\infty}^x f(x)dx$$

$$f(x) = \frac{dF(x)}{dx}$$

Example:

The random duration of some components is a variable whose density function is $f(x) = 0,10e^{-0,10x} \quad \forall x \geq 0$. Calculate F(X).

$$F(x) = \int_{-\infty}^x f(x)dx = \int_{-\infty}^x 0,10e^{-0,10x}dx = 1 - e^{-0,10x} \quad \forall x \geq 0$$



f(x) Properties

1. As $F(x)$ is non – decreasing $\forall x \in \mathbb{R}$

$$f(x) = \frac{dF(x)}{dx} \quad \text{is positive} \quad \forall x \in \mathbb{R}$$

$$2. \int_{-\infty}^{+\infty} f(x)dx = \lim_{x \rightarrow +\infty} F(x) - \lim_{x \rightarrow -\infty} F(x) = 1 - 0 = 1$$

$$3. f(x) = \frac{dF(x)}{dx} = \lim_{\Delta x \rightarrow 0} \frac{F(x+\Delta x) - F(x)}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{P(x < X \leq x + \Delta x)}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{P(\Delta x)}{\Delta x}$$

$f(x)$ is the limit of the mass of probability of an interval divided by the length of the said interval, that is to say, is a probability density, hence its name of the probability density function.

The probability of a specific value x is zero, but around that point there is a probability density $f(x)$ and the probability of a differential range dx around x is $P(dx) = f(x)dx$

Expected Value of a continuous variable

$$E(\mathbf{X}) = m = \mu = \int_{-\infty}^{+\infty} xf(x)$$

Examples:

Find the expected value of the following random variables:

- Variable uniformly distributed between 0 and 1
- Exponential distribution of parameter $\alpha = 0.01$

Properties

If $g(\mathbf{X})$ is a delimited function of x

$$E(g(\mathbf{X})) = \int_x g(x)f(x)dx \quad \text{if } \mathbf{X} \text{ is continuous}$$

If $g(\mathbf{X}) = \mathbf{X}$ then we have the particular case of the average of $\mathbf{X}$

Median

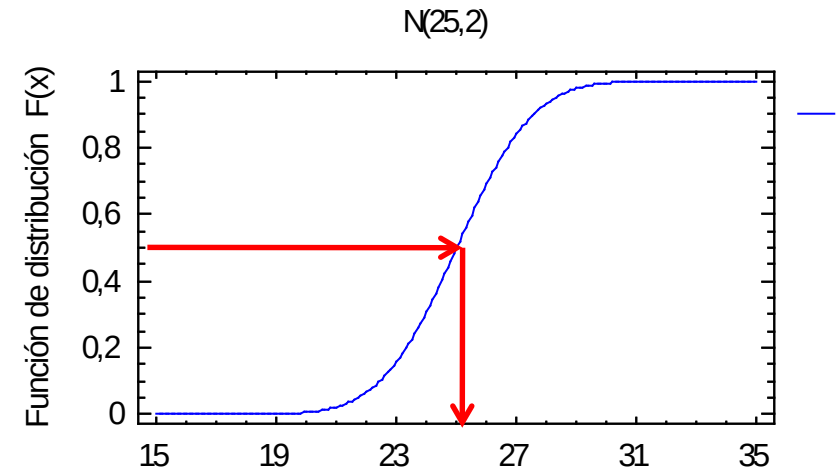
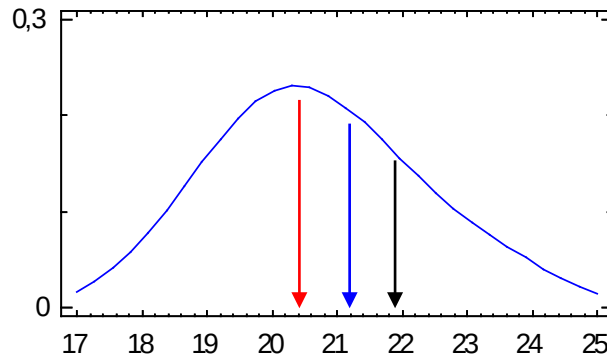
Value that leaves to its left the 50% of the values of the variable. It corresponds with the percentile 50 or P_{50}

The median is the value a such that $F(a)=0,5$

Mode:

Value that presents a maximum relative in:

- $f(x)$ (continuous variables)



Concept

Measure of dispersion around the average of the variable

$$\sigma^2(X) = D^2(X) = E(X-m)^2$$

Notes:

$$E(x-m)=0$$

($D^2(X)$ means variance of X)

Calculation:

$$\sigma^2(X) = D^2(X) = \int_x (x - m)^2 f(x) dx$$

Standard Deviation

$$\sigma = +\sqrt{\sigma^2}$$

$$D(X) = +\sqrt{D^2(X)}$$

Coefficient of variation

$$CV = \frac{\sigma}{m} 100 \quad (\text{dimensionless})$$

Main distributions of probability

- *DISCRETE VARIABLES*

- Bernouilli Distribution
- Hypergeometric Distribution
- Binomial Distribution
- Negative Binomial Distribution
- Poisson Distribution
- Distribución multinomial

- *CONTINUOUS VARIABLES*

- Uniform Distribution
- Exponential Distribution
- Normal Distribution
- X^2 -Distribution
- t-Distribution
- F-Distribution



Uniform Distribution

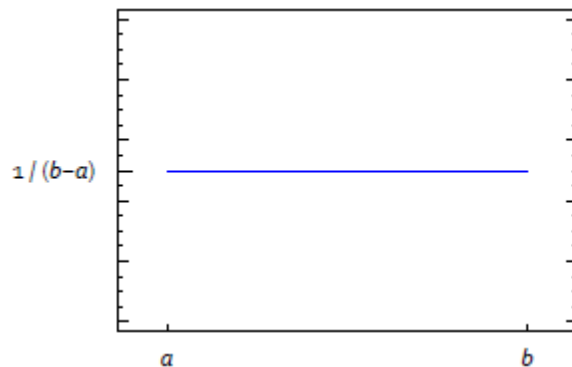
The **uniform distribution** appears in contexts where the probability is distributed "by equal" in all a continuous range of values possible.

Within the sample space, the probability of an interval will be proportional to the "width" of the same.

Density function $f(x)$

$$f = \frac{1}{b-a} \quad \text{if } a \leq x \leq b$$

0 *in the rest*

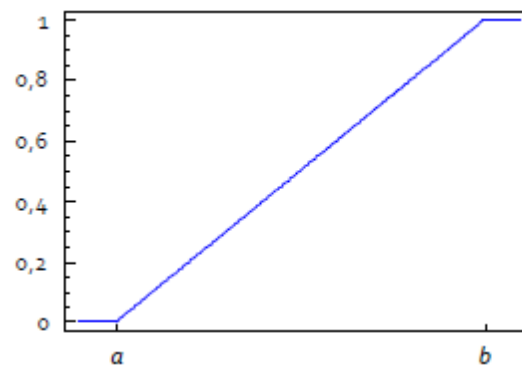


Distribution function $F(x)$

$$F(x) = \frac{x-a}{b-a} \quad \text{if } a \leq x \leq b$$

0 *if* $x \leq a$

1 *si* $b \leq x$



Uniform Distribution

Notation: $X \sim U(a, b)$ $[a, b]$ range of possible values of X

Mean: $E(x) = \frac{a+b}{2}$

Variance: $D^2(x) = \frac{(b-a)^2}{12}$

Examples:

- Choose randomly a point in the road that joins Valencia and Madrid.
- Instant of a day in which it is activated an alarm that it is damaged and is activated completely randomly once every day.

Exercise 1:

The time that it takes a particular server to respond to a request is uniformly distributed between 1 and 5 seconds. Calculates the probability of

- ... the server take less than 2 seconds in responding to a request.
- ... take more than 3.5 seconds to respond to a request.
- ... take more than 7 seconds.
- ... take in answer to a request between 2 and 3.5 seconds.
- ... take exactly 3 seconds.

Uniform Distribution

Exercise 2:

In a factory it is known that the length of parts supplied by a particular supplier varies uniformly between 500 and 505 mm. Only be considered valid the pieces with a length of 503 ± 1 mm.

- a) what is the probability that a piece of that provider is valid?
- b) what is the expected average length and the standard deviation of the parts provided by this provider?

Exercise 3:

The van of security providing euros and currencies to a particular bank branch visits every day this branch in a totally random moment between 8 am and 3 pm, in order to prevent possible theft. What is the instant of day that a thief should choose to visit the branch if he wants to maximize the probability of coincide with the visit of the van?

Exponential Distribution

The **exponential distribution** is used with some frequency for modelling the **time between occurrences or events**, as troubleshooting, visits, etc., or the **time until the failed** (time of life) of a given component, system or product.

It is assumed that the **average rate of occurrences** per unit of time (or of space, etc.) is **constant**.

It is usually to say that **exponential distribution "has not memory"**: the time elapsed since the last occurrence not influences in the calculation of the probability about the next occurrence.

In the particular case of modelling the life time of a product, exponential distribution is adequate when it is considered that **there is not wear in the product and that the failures is produce only of accidental mode** (there are not early failures due to errors of manufacturing or failures due to the fatigue or wear of the material).

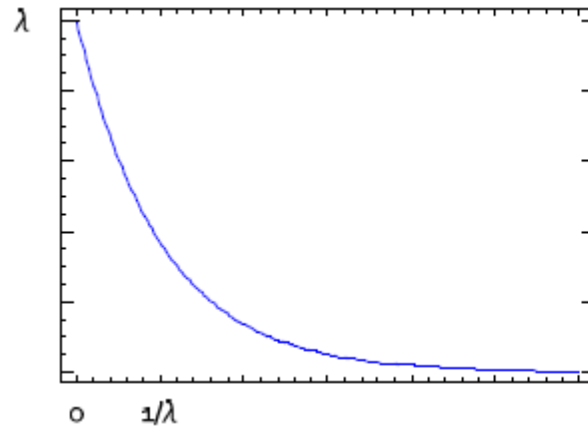
In certain mode, **the exponential distribution and Poisson distribution are complementary**: both depart from the same hypothesis; one (the exponential) measures the time between occurrences, and the other (Poisson) measured the occurrences per time unit. In both cases, the parameter λ represents the constant average rate of occurrences per unit of time.

It is what is called a **homogeneous Poisson process**.

Exponential Distribution

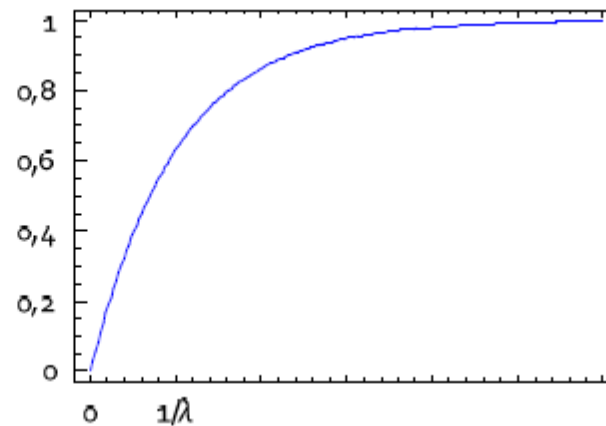
Density function $f(x)$

$$f = \lambda e^{-\lambda x} \quad \text{si } x \geq 0$$



Distribution function $F(x)$

$$F(x) = P(X \leq x) = 1 - e^{-\lambda x} \quad \text{si } x \geq 0$$



λ average rate of occurrences per time unit

Examples: In any of the examples of the Poisson distribution, considering as a random variable of interest the time until the next event, rather than look at how many events occur in a given space of time

Exponential Distribution

Notation: $X \sim \text{Exp}(\lambda)$ λ average rate of occurrences per time unit

Mean: $E(x) = \frac{1}{\lambda}$

Variance: $D^2(x) = \frac{1}{\lambda^2}$

Exercise 4 :

It has studied the behavior of the life time of a type of circuits and is known that follows a exponential distribution of average 1,000 hours.

What is the probability that one of these circuits work still, 500 hours after his release up?

Exercise 5 :

It is known that the queue time in the bank windows of a bank branches behaves like an exponential variable, although the rate of arrivals λ is different in each branch. If in the branch of adjoining street the probability of waiting more than 4-minute in the bank window is 45%, what is average waiting time (average queue time) in that branch?

Exercise 6 :

Given a random variable $X \sim \text{Exp}(\lambda=20)$, being μ its mean and σ its standard deviation. What is the probability that a specific observation of X falls in the interval $\mu \pm 3\sigma$?

Exponential Distribution

Remember that the exponential distribution "does not have memory".

Exercise 7:

The life time of certain electronic components is distributed exponentially, with average 10,000 hours. It is known that one of these components has been running already 5,000 hours. What is the probability of that its total life time exceed the 11,000 hours?

Exercise 8:

The queue time in the queue of a given cinema follows a exponential distribution of mean 5 minutes.

- a) what is the probability of having to wait more than 4 minutes in the queue, if we just reach?
- b) what is the probability that, counting from now, we have to wait more than 4 minutes to be served, if we have been in the queue 10 min?

Reliability

Reliability of a device, an appliance, etc.: **probability that work**, at a given time

Often it is used the notation R_A to express the reliability of the component or of the system A; It means:

$$R_A = P(A \text{ works})$$

Example:

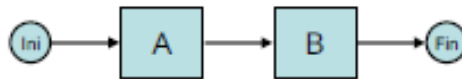
The reliability of a switch to the 10000 operating hours is the probability that work, after 10000 hours since its installation or start up.

Exercise 9 :

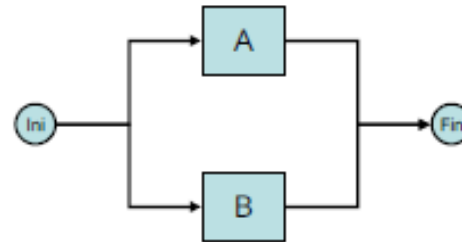
The reliability of a device to the 5000 operating hours, is less, equal, or greater than the same device to the 7,000 hours of operation? Why?

Reliability

Reliability of systems in series and in parallel. Often, we will be interested in calculating the reliability of systems (computer networks, circuits, etc.) in series, in parallel, or in combined forms of those two basic distributions.



Esquema de un sistema formado por dos componentes conectados **en serie**



Esquema de un sistema formado por dos componentes conectados **en paralelo**

If the components are independent, then:

$$R_{sist} = R_A R_B$$

$$R_{sist} = 1 - (1 - R_A)(1 - R_B) \quad (*)$$

serial system

parallel sytem

Reliability

(*) Probability of the Union of events (independent)

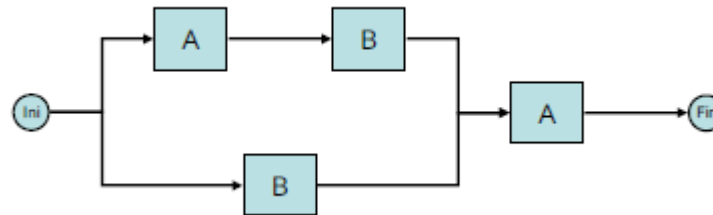
To calculate the probability of the Union of two or more independent events, we have an alternative and equivalent to the general formula (and, in general, recommended): use the complementary event property (and the laws of Morgan).

$$\begin{aligned} P(A \cup B \cup C) &= [\text{complementary event}] = 1 - P(\overline{A \cup B \cup C}) = [\text{Morgan}] \\ &= 1 - P(\bar{A} \cap \bar{B} \cap \bar{C}) = [\text{independent events}] = 1 - P(\bar{A})P(\bar{B})P(\bar{C}) \\ &= [\text{complementary event}] = 1 - (1 - P(A))(1 - P(B))(1 - P(C)) \end{aligned}$$

Reliability

Exercise 10:

A system is made up of components of types A and B, the reliability of which in the 5,000 hours is 60 and 80%, respectively. The design of the system is shown in the figure.



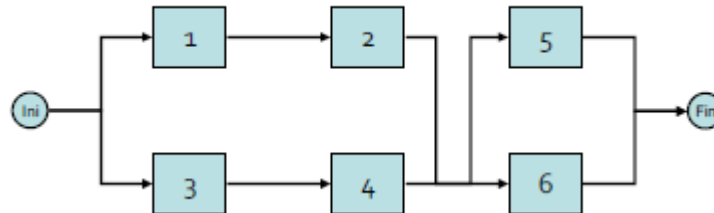
Calculate the reliability of the system when you get 5,000 hours of operation; assume that the components behave independently between them.

Reliability

Exercise 11:

The following figure represents a network of processors where each processor has an average life time of 21827 hours.

What is the probability that the network **does not work**, after 20,000 working hours? We can assume that the processors operate independently.



Normal Distribution

The **normal distribution** is one of more used continuous distributions to represent the behavior of random quantities, since in the reality many variable random is in accordance to this model.

Is a "**symmetrical**" **model**: the probability is concentrated in a central area of values, and is decreasing symmetrically according to is away of such central area central. Remember a bell.

NOT it is the only symmetrical distribution, but it is that arises in the practice to the observe and study of random phenomena in the natural, social, etc. sciences.

Typical examples of random variables. that follow a normal distribution are the height and weight of a population.

Normal Distribution

Density Function $f(x)$

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

- μ : *position parameter.*
- σ : *scattering parameter.*

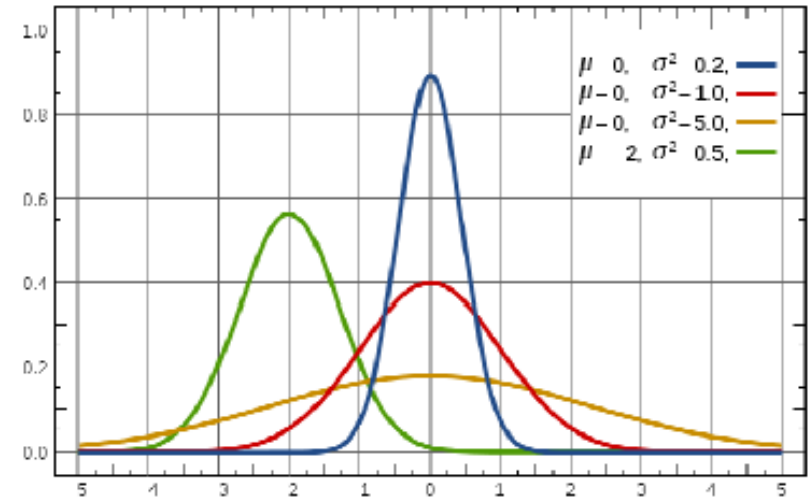
Distribution Function $F(x)$

$$F(x) = P(X \leq x) = \text{*THERE ISN'T ANY ANALYTIC EXPRESSION*}$$

Notation: $X \sim N(\mu, \sigma)$

Mean: $E(x) = \mu$

Variance: $D^2(x) = \sigma^2$



Normal Distribution

We will see that we will have to go to a **table** to calculate probabilities, but we can highlight the following results. If $X \sim N(\mu, \sigma)$,

then:

- $P(\mu - \sigma \leq X \leq \mu + \sigma) \approx 0,6827 \approx 2/3$
- $P(\mu - 2\sigma \leq X \leq \mu + 2\sigma) \approx 0,9545 \approx 95\%$
- $P(\mu - 3\sigma \leq X \leq \mu + 3\sigma) \approx 0,9973 \approx 99,7\%$

Exercise 9:

The weight of boxes of printed circuit manufactured by a certain company behaves according to a normal distribution of mean 5 Kg and standard deviation 0.1 Kg. in which range of values is expected to be included the weight of 95% of the boxes produced by this company?

Exercise 10:

99.7% of individuals in a given population have a height between 1.51 and 1.93 m, being the average of 1,72m. Assuming that this variable follow a normal distribution, estimated the value of the standard deviation of the stature of this population.

Normal Distribution

Linear combination of normals

$$X_i \sim N(\mu_i, \sigma_i), \quad i = 1, \dots, n, \quad \text{and independent,}$$

$$\text{and} \quad Y = c_0 + c_1 X_1 + \dots + c_n X_n \rightarrow Y \sim N(\mu, \sigma),$$

$$\text{with} \quad \mu = c_0 + \sum c_i \mu_i \quad \text{and} \quad \sigma^2 = \sum c_i^2 \sigma_i^2$$

Particular case

$$X_i \sim N(\mu_i, \sigma_i), \quad i = 1, 2 \quad \text{and independent,}$$

$$Y = X_1 + X_2 \rightarrow Y \sim N(\mu, \sigma), \quad \text{with} \quad \mu = \mu_1 + \mu_2 \quad \text{and} \quad \sigma^2 = \sigma_1^2 + \sigma_2^2$$

$$Y = X_1 - X_2 \rightarrow Y \sim N(\mu, \sigma), \quad \text{with} \quad \mu = \mu_1 - \mu_2 \quad \text{and} \quad \sigma^2 = \sigma_1^2 + \sigma_2^2$$

Normal Distribution

Standard Normal Distribution

Normal distribution where the mean is 0 and the variance (and standard deviation) is set to 1.

$$Z \sim N(0, 1)$$

$$X \sim N(\mu, \sigma) \Rightarrow Z = \frac{X - \mu}{\sigma} \sim N(0, 1)$$

The advantage of standardize is that probabilities related to distribution $N(0,1)$ are collected in a table (see form of the subject).

To the value of the $N(0,1)$ that has a probability α of be exceeded is called z_α

DISTRIBUCIÓN NORMAL TIPIFICADA



$$\Phi(z) = P(Z \leq z) = \int_{-\infty}^z \frac{1}{\sqrt{2\pi}} \cdot e^{-\frac{t^2}{2}} dt$$



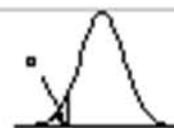
z	0	1	2	3	4	5	6	7	8	9	z	0	1	2	3	4	5	6	7	8	9
-3.0	0.0044	0.0043	0.0042	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036	0.0035	0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
-2.9	0.0044	0.0043	0.0042	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036	0.0035	0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
-2.8	0.0044	0.0043	0.0042	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036	0.0035	0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
-2.7	0.0044	0.0043	0.0042	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036	0.0035	0.3	0.6179	0.6217	0.6256	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
-2.6	0.0044	0.0043	0.0042	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036	0.0035	0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
-2.5	0.0044	0.0043	0.0042	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036	0.0035	0.5	0.6896	0.6930	0.6965	0.7000	0.7034	0.7068	0.7102	0.7135	0.7169	0.7202
-2.4	0.0044	0.0043	0.0042	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036	0.0035	0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
-2.3	0.0044	0.0043	0.0042	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036	0.0035	0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
-2.2	0.0044	0.0043	0.0042	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036	0.0035	0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
-2.1	0.0044	0.0043	0.0042	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036	0.0035	0.9	0.8150	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8366	0.8390
-2.0	0.0044	0.0043	0.0042	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036	0.0035	1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
-1.9	0.0044	0.0043	0.0042	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036	0.0035	1.1	0.8643	0.8666	0.8688	0.8709	0.8729	0.8749	0.8769	0.8788	0.8807	0.8826
-1.8	0.0044	0.0043	0.0042	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036	0.0035	1.2	0.8840	0.8860	0.8879	0.8897	0.8915	0.8932	0.8949	0.8966	0.8982	0.8998
-1.7	0.0044	0.0043	0.0042	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036	0.0035	1.3	0.9012	0.9030	0.9048	0.9065	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162
-1.6	0.0044	0.0043	0.0042	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036	0.0035	1.4	0.9122	0.9139	0.9155	0.9172	0.9188	0.9204	0.9219	0.9235	0.9250	0.9265
-1.5	0.0044	0.0043	0.0042	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036	0.0035	1.5	0.9232	0.9248	0.9264	0.9279	0.9294	0.9309	0.9324	0.9338	0.9353	0.9367
-1.4	0.0044	0.0043	0.0042	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036	0.0035	1.6	0.9362	0.9376	0.9389	0.9401	0.9413	0.9425	0.9437	0.9448	0.9459	0.9470
-1.3	0.0044	0.0043	0.0042	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036	0.0035	1.7	0.9454	0.9465	0.9475	0.9485	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
-1.2	0.0044	0.0043	0.0042	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036	0.0035	1.8	0.9464	0.9474	0.9483	0.9492	0.9501	0.9510	0.9519	0.9528	0.9537	0.9545
-1.1	0.0044	0.0043	0.0042	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036	0.0035	1.9	0.9473	0.9482	0.9491	0.9499	0.9508	0.9516	0.9524	0.9532	0.9540	0.9548
-1.0	0.0044	0.0043	0.0042	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036	0.0035	2.0	0.9472	0.9480	0.9488	0.9496	0.9504	0.9512	0.9519	0.9527	0.9534	0.9542
-0.9	0.0044	0.0043	0.0042	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036	0.0035	2.1	0.9481	0.9488	0.9495	0.9502	0.9509	0.9516	0.9522	0.9529	0.9535	0.9542
-0.8	0.0044	0.0043	0.0042	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036	0.0035	2.2	0.9486	0.9492	0.9498	0.9504	0.9510	0.9516	0.9521	0.9527	0.9532	0.9538

$$P(z < 1.75) = \Phi(1.75) = 0.9599$$

DISTRIBUCIÓN NORMAL TIPIFICADA



$$\Phi(z) = P(Z \leq z) = \int_{-\infty}^z \frac{1}{\sqrt{2\pi}} \cdot e^{-\frac{t^2}{2}} dt$$



z	0	1	2	3	4	5	6	7	8	9	z	0	1	2	3	4	5	6	7	8	9
-3	0,0013	0,0010	0,0007	0,0005	0,0003	0,0002	0,0002	0,0001	0,0001	0,0000	0,0	0,5000	0,5040	0,5080	0,5120	0,5160	0,5199	0,5239	0,5279	0,5319	0,5359
-2,9	0,0010	0,0008	0,0006	0,0004	0,0003	0,0002	0,0001	0,0001	0,0000	0,0000	0,1	0,5398	0,5438	0,5478	0,5517	0,5557	0,5596	0,5636	0,5675	0,5714	0,5753
-2,8	0,0008	0,0006	0,0004	0,0003	0,0002	0,0001	0,0001	0,0000	0,0000	0,0000	0,2	0,5793	0,5832	0,5871	0,5910	0,5948	0,5987	0,6026	0,6064	0,6103	0,6141
-2,7	0,0006	0,0004	0,0003	0,0002	0,0001	0,0001	0,0000	0,0000	0,0000	0,0000	0,3	0,6179	0,6217	0,6256	0,6293	0,6331	0,6368	0,6406	0,6443	0,6480	0,6517
-2,6	0,0004	0,0003	0,0002	0,0001	0,0001	0,0000	0,0000	0,0000	0,0000	0,0000	0,4	0,6554	0,6591	0,6628	0,6664	0,6700	0,6736	0,6772	0,6808	0,6844	0,6879
-2,5	0,0003	0,0002	0,0001	0,0001	0,0000	0,0000	0,0000	0,0000	0,0000	0,0000	0,5	0,6895	0,6930	0,6965	0,7000	0,7034	0,7068	0,7102	0,7135	0,7169	0,7203
-2,4	0,0002	0,0001	0,0001	0,0000	0,0000	0,0000	0,0000	0,0000	0,0000	0,0000	0,6	0,7257	0,7291	0,7324	0,7357	0,7389	0,7422	0,7454	0,7486	0,7517	0,7549
-2,3	0,0001	0,0001	0,0000	0,0000	0,0000	0,0000	0,0000	0,0000	0,0000	0,0000	0,7	0,7580	0,7611	0,7642	0,7673	0,7704	0,7734	0,7764	0,7794	0,7823	0,7852
-2,2	0,0001	0,0000	0,0000	0,0000	0,0000	0,0000	0,0000	0,0000	0,0000	0,0000	0,8	0,7881	0,7910	0,7939	0,7967	0,7995	0,8023	0,8051	0,8078	0,8106	0,8133
-2,1	0,0000	0,0000	0,0000	0,0000	0,0000	0,0000	0,0000	0,0000	0,0000	0,0000	0,9	0,8190	0,8218	0,8246	0,8273	0,8300	0,8327	0,8354	0,8381	0,8408	0,8435
-2,0	0,0000	0,0000	0,0000	0,0000	0,0000	0,0000	0,0000	0,0000	0,0000	0,0000	1,0	0,8413	0,8438	0,8461	0,8486	0,8508	0,8531	0,8554	0,8577	0,8599	0,8621
-1,9	0,0000	0,0000	0,0000	0,0000	0,0000	0,0000	0,0000	0,0000	0,0000	0,0000	1,1	0,8643	0,8666	0,8686	0,8708	0,8729	0,8749	0,8770	0,8790	0,8810	0,8830
-1,8	0,0000	0,0000	0,0000	0,0000	0,0000	0,0000	0,0000	0,0000	0,0000	0,0000	1,2	0,8840	0,8860	0,8879	0,8897	0,8915	0,8934	0,8952	0,8970	0,8987	0,9005
-1,7	0,0000	0,0000	0,0000	0,0000	0,0000	0,0000	0,0000	0,0000	0,0000	0,0000	1,3	0,9032	0,9049	0,9066	0,9082	0,9099	0,9115	0,9131	0,9147	0,9162	0,9177
-1,6	0,0000	0,0000	0,0000	0,0000	0,0000	0,0000	0,0000	0,0000	0,0000	0,0000	1,4	0,9192	0,9207	0,9222	0,9236	0,9251	0,9266	0,9280	0,9294	0,9308	0,9321
-1,5	0,0000	0,0000	0,0000	0,0000	0,0000	0,0000	0,0000	0,0000	0,0000	0,0000	1,5	0,9332	0,9346	0,9359	0,9374	0,9388	0,9398	0,9413	0,9427	0,9441	0,9455
-1,4	0,0000	0,0000	0,0000	0,0000	0,0000	0,0000	0,0000	0,0000	0,0000	0,0000	1,6	0,9452	0,9465	0,9477	0,9489	0,9499	0,9509	0,9519	0,9528	0,9537	0,9546
-1,3	0,0000	0,0000	0,0000	0,0000	0,0000	0,0000	0,0000	0,0000	0,0000	0,0000	1,7	0,9554	0,9564	0,9573	0,9582	0,9591	0,9599	0,9608	0,9616	0,9625	0,9633
-1,2	0,0000	0,0000	0,0000	0,0000	0,0000	0,0000	0,0000	0,0000	0,0000	0,0000	1,8	0,9641	0,9649	0,9656	0,9664	0,9671	0,9678	0,9686	0,9693	0,9699	0,9706
-1,1	0,0000	0,0000	0,0000	0,0000	0,0000	0,0000	0,0000	0,0000	0,0000	0,0000	1,9	0,9713	0,9719	0,9726	0,9732	0,9738	0,9744	0,9750	0,9756	0,9761	0,9767
-1,0	0,0000	0,0000	0,0000	0,0000	0,0000	0,0000	0,0000	0,0000	0,0000	0,0000	2,0	0,9772	0,9778	0,9783	0,9788	0,9793	0,9798	0,9803	0,9808	0,9812	0,9817
-0,9	0,0000	0,0000	0,0000	0,0000	0,0000	0,0000	0,0000	0,0000	0,0000	0,0000	2,1	0,9821	0,9826	0,9830	0,9834	0,9838	0,9842	0,9846	0,9850	0,9854	0,9857
-0,8	0,0000	0,0000	0,0000	0,0000	0,0000	0,0000	0,0000	0,0000	0,0000	0,0000	2,2	0,9861	0,9864	0,9868	0,9871	0,9875	0,9878	0,9881	0,9884	0,9887	0,9890

$$Z_{0,025} = 1,96$$

$$1 - 0,025 = 0,975$$

Exercise 11:

The value of the resistors manufactured by a certain company is distributed randomly following a normal law with an average of 9.76 ohms and standard deviation of 0.05 ohms. All resistance with a value below 9.69 ohms are considered defective.

- what is the probability that a resistor is defective?
- if we take 5 resistors randomly what is the probability of having at least 2 defective?

Ejercicio 12 :

The CPU time that a kind of programs need to be run on a server is distributed in a normal manner, with a standard deviation of 2 minutes.

Calculates the average time of CPU consumed by programs of this kind, if you know that 20% of them need more than 15 minutes to run.

Exercise 13:

A company distributes packs of smartphones plus speakers for such phones. Weight of the phones is distributed according to a normal of average 135 g and standard deviation of 0.7 g, while the weight of the speakers follow also a normal distribution of average 1 Kg and standard deviation 3 g.

Calculate the probability of that one of these packs (before packing) have a total weight above to 1,140 Kg.

Exercise 14:

Average time necessary to perform a certain type of work in a company of graphic design is of 20 minutes, being the standard deviation of 3 minutes. At the same time that it is started the work is called to the messenger that has that collect it and deliver it to the client. The messenger takes to reach the design company an average of 22 minutes, with a standard deviation of 5 minutes. Assuming normality and independence of variables, calculate the probability that the messenger reach the company before it is completed the work that must be collected.

Exercise 15:

A company manufactures cylindrical parts of plastic and enter them in individual boxes to be distributed. The length of the parts and of the boxes are $N(15 \text{ cm}, 1 \text{ cm})$ and $N(18 \text{ cm}, 1 \text{ cm})$, respectively. The width of the boxes is always superior to the parts (as include space for a padded of internal protection). What is the probability of that a piece do not fit in its box, due to its length?

Exercise 16:

Weight of some certain packages is normal distributed with average 15 Kg and standard deviation 2 Kg.

a. what is the probability of that the weight of one of these packages is below to 13 Kg?

If we take 10 packages randomly

b. what is the probability of that at most 1 of them is below to 13 Kg?

c. what is the probability that the total weight of 10 packages does not exceed 130 Kg?

Central Limit Theorem

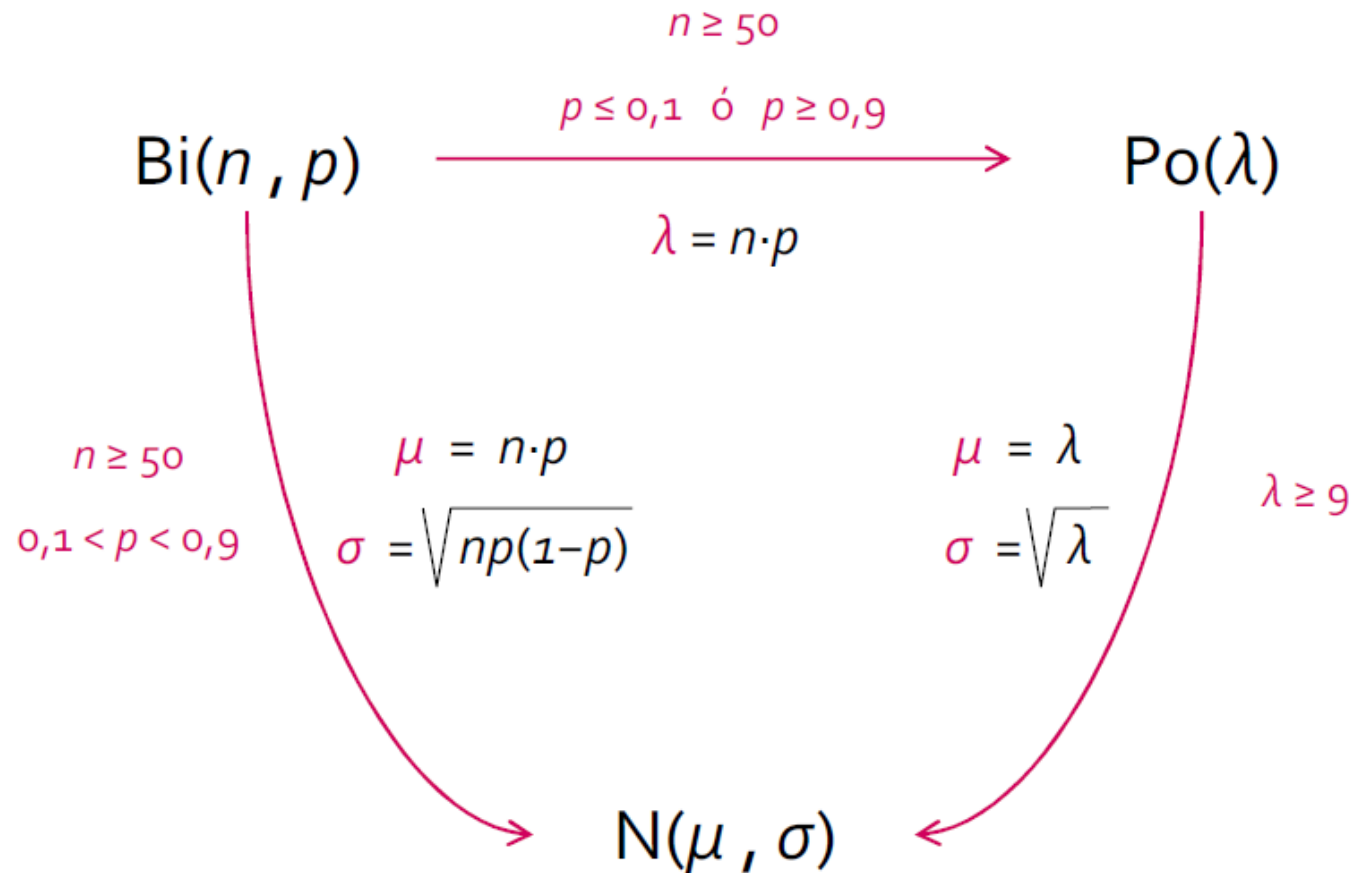
As already mentioned, sometimes it is possible to approximate calculations of other distributions using the normal distribution. This is due to the Central Limit Theorem.

- **Central limit theorem (CLT)**: the sum of independent random variable and with the same distribution tends to be distributed according to a normal distribution.

Thus, we can deduce that the sum of uniforms, the sum of exponential, the sum of binomial... distributes approximately normally (with a mean equal to the sum of the mean and variance equal to the sum of the variances).

Also thanks to the CLT it can be demonstrated that the binomial and poisson distributions can be approached to a normal distribution.

Central Limit Theorem



Exercise 17:

The average number of calls that are received at a hospital switchboard is constant and equal to 4 per minute. Calculates the probability that in the next hour more than 225 calls are received.

Exercise 18:

Calculates the probability of rolling 120 times a die (not loaded), get a 6 in at least 20% of the rolls.

Exercise 19:

40% of the stickers made by a specific company are defective. If the stickers are present in leaves of 60 units, what is the probability of having in a sheet more than the half of the stickers defective?

χ_n^2 Distribution

Defines the χ_n^2 as the sum of the squares of n independent standard normal variables

Mean and Variance::

$$E(\chi_n^2) = n \quad D^2(\chi_n^2) = 2n$$

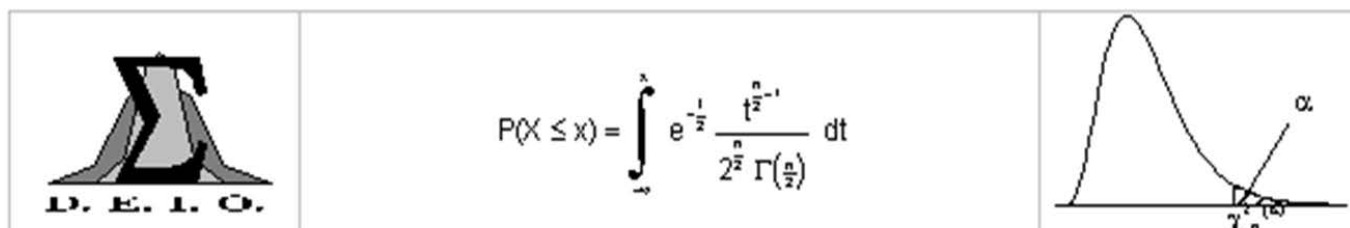
Their probabilities are obtained in the tables

Among other uses is used:

1. in the comparison of averages
2. in the test on equality of variances
3. as a criterion for decision in the Pearson test

DISTRIBUCIÓN χ^2

DISTRIBUCIÓN CHI-CUADRADO DE PEARSON



n	0.9995	0.999	0.995	0.99	0.975	0.95	0.90	0.50	0.10	0.050	0.025	0.01	0.005	0.001	0.0005
1	0.000	0.000	0.000	0.000	0.001	0.004	0.016	0.455	2.706	3.842	5.024	6.635	7.879	10.827	12.115
2	0.001	0.002	0.010	0.020	0.051	0.103	0.211	1.386	4.605	5.992	7.378	9.210	10.597	13.815	15.201
3	0.015	0.024	0.072	0.115	0.216	0.352	0.584	2.366	6.251	7.815	9.348	11.345	12.838	16.266	17.731
4	0.064	0.091	0.207	0.297	0.484	0.711	1.064	3.357	7.779	9.488	11.143	13.277	14.860	18.466	19.998
5	0.158	0.210	0.412	0.554	0.831	1.146	1.610	4.352	9.236	11.071	12.833	15.086	16.750	20.515	22.106
6	0.299	0.381	0.676	0.872	1.237	1.635	2.204	5.348	10.595	12.592	14.449	16.812	18.548	22.467	24.102
7	0.485	0.599	0.989	1.239	1.690	2.167	2.833	6.346	12.017	14.067	16.013	18.475	20.278	24.321	26.018
8	0.710	0.857	1.344	1.647	2.180	2.733	3.490	7.344	13.362	15.507	17.535	20.090	21.955	26.124	27.867
9	0.972	1.156	1.735	2.179	2.700	3.325	4.168	8.343	14.684	16.919	19.023	21.666	23.589	27.877	29.667
10	1.259	1.479	2.150	2.558	3.177	3.571	4.591	9.348	15.987	18.307	20.483	23.209	25.188	29.588	31.410
11	1.587	1.834	2.603	3.054	3.816	4.575	5.578	10.341	17.275	19.675	21.920	24.725	26.757	31.264	33.138
12	1.905	2.214	3.074	3.571	4.404	5.226	6.304	11.340	18.549	21.026	23.337	26.217	28.300	32.909	34.821
13	2.205	2.617	3.565	4.107	5.009	5.892	7.042	12.340	19.812	22.362	24.736	27.688	29.819	34.527	36.477

$$X = \chi^2_9$$

$$P(X > 2.7) = 0.975$$

$$\chi^2_{9,0.10} = 14.684$$

t-Student Distribution

It is defined as the ratio of a standard normal and the square root of a Chi-square divided by its degrees of freedom, both independent

$$t_n = \frac{Z}{\sqrt{\frac{\chi^2}{n}}}$$

$$P(t > t_n^{\alpha/2}) = \alpha/2$$

$$P(t > |t_n^{\alpha/2}|) = \alpha$$

Mean and variance::

$$E(t_n) = 0$$

$$D^2(t_n) = n/(n-2)$$

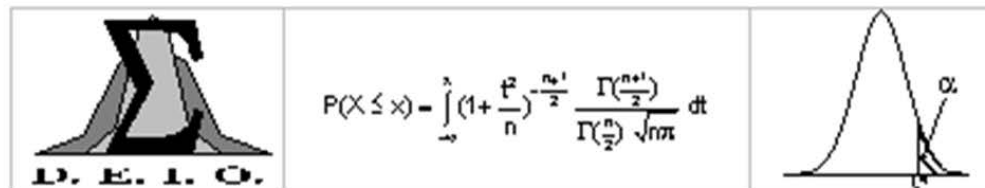
Their probabilities are obtained in the tables

Among other uses is used:

1. in the comparison of averages
2. in the test of variance
3. as a criterion for decision in the Pearson test

DISTRIBUCIÓN t de STUDENT

DISTRIBUCIÓN t de Student



Probabilidad de una cola													
n	0.0005	0.001	0.005	0.01	0.025	0.05	0.1	0.2	0.25	0.3	0.4	0.45	0.475
1	636.578	318.289	63.656	31.821	12.706	6.314	3.078	1.376	1.000	0.727	0.325	0.158	0.079
2	31.600	22.328	9.925	6.965	4.303	2.920	1.886	1.061	0.816	0.617	0.289	0.142	0.071
3	12.924	10.214	5.841	4.541	3.182	2.353	1.638	0.978	0.765	0.584	0.277	0.137	0.068
4	8.610	7.173	4.604	3.747	2.776	2.132	1.533	0.941	0.741	0.569	0.271	0.134	0.067
5	6.869	5.894	4.032	3.365	2.571	2.015	1.476	0.920	0.727	0.559	0.267	0.132	0.066
6	5.959	5.208	3.707	3.143	2.447	1.943	1.440	0.906	0.718	0.553	0.265	0.131	0.065
7	5.408	4.785	3.499	2.958	2.365	1.895	1.415	0.896	0.711	0.549	0.263	0.130	0.065
8	5.041	4.501	3.355	2.896	2.306	1.860	1.397	0.889	0.706	0.546	0.262	0.130	0.065
9	4.781	4.297	3.250	2.821	2.262	1.833	1.383	0.883	0.703	0.543	0.261	0.129	0.064
10	4.587	4.144	3.169	2.764	2.228	1.812	1.372	0.879	0.700	0.542	0.260	0.129	0.064

$$X = t_6$$

$$P(X > 1.44) = 0.10$$

$$t_{6,0.01} = 3.143$$

F-Snedecor Distribution

Is defined as the ratio of two chi square independent, divided each by its degrees of freedom

$$F_{n_1, n_2} = \frac{\chi_{n_1}^2 / n_1}{\chi_{n_2}^2 / n_2}$$

Their probabilities are obtained in the tables

Among other uses is used:

1. in the comparison of averages
2. in the test of variance
3. as a criterion for decision in the Pearson test

DISTRIBUCIÓN F de Snedecor ($F_{n_1, n_2}^{(p)}$)

+

	Grados de libertad de la varianza mayor															
	1		2		3		4		5		6		7		8	
p→	0.05	0.01	0.05	0.01	0.05	0.01	0.05	0.01	0.05	0.01	0.05	0.01	0.05	0.01	0.05	0.01
1	161.45	4052.2	199.50	4999.3	215.71	540.65	224.58	5624.3	230.16	5763.9	233.99	5858.9	236.77	5928.3	238.88	5980.9
2	18.51	98.50	19.00	99.00	19.16	99.16	19.25	99.25	19.30	99.30	19.33	99.33	19.35	99.36	19.37	99.38
3	10.13	34.12	9.55	30.82	9.28	29.46	9.12	28.71	9.01	28.24	8.94	27.91	8.89	27.67	8.85	27.49
4	7.71	21.20	6.94	18.00	6.59	16.69	6.39	15.98	6.26	15.52	6.16	15.21	6.09	14.98	6.04	14.80
5	6.61	16.26	5.79	13.27	5.41	12.66	5.19	11.39	5.05	10.97	4.95	10.67	4.88	10.46	4.82	10.29
6	5.99	13.75	5.14	10.92	4.76	9.78	4.53	9.15	4.39	8.75	4.28	8.47	4.21	8.26	4.15	8.10
7	5.59	12.25	4.74	9.55	4.35	8.46	4.12	7.85	3.97	7.46	3.87	7.19	3.79	6.99	3.73	6.84
8	5.32	11.26	4.46	8.65	4.07	7.59	3.84	7.01	3.69	6.63	3.58	6.37	3.50	6.18	3.44	6.03
9	5.12	10.56	4.26	8.02	3.86	6.99	3.63	6.42	3.48	6.06	3.37	5.80	3.29	5.61	3.23	5.47
10	4.96	10.04	4.10	7.59	3.70	6.55	3.48	5.99	3.33	5.64	3.22	5.39	3.14	5.20	3.07	5.06
11	4.84	9.65	3.98	7.21	3.59	6.28	3.36	5.67	3.20	5.32	3.09	5.07	3.01	4.89	2.95	4.74
12	4.75	9.33	3.89	6.93	3.49	5.95	3.26	5.41	3.11	5.06	3.00	4.82	2.91	4.64	2.85	4.50
13	4.67	9.07	3.81	6.70	3.41	5.74	3.18	5.21	3.03	4.86	2.92	4.62	2.83	4.44	2.77	4.30
14	4.60	8.86	3.74	6.51	3.34	5.56	3.11	5.04	2.96	4.69	2.85	4.46	2.76	4.28	2.70	4.14
15	4.54	8.68	3.68	6.36	3.29	5.42	3.06	4.89	2.90	4.56	2.79	4.32	2.71	4.14	2.64	4.00
16	4.49	8.53	3.63	6.23	3.24	5.29	3.01	4.77	2.85	4.44	2.74	4.20	2.66	4.03	2.59	3.89
17	4.45	8.40	3.59	6.11	3.20	5.19	2.96	4.67	2.81	4.34	2.70	4.10	2.61	3.93	2.55	3.76

$$X \equiv F_{3,10}$$

$$P(X > 6.6) = 0.01$$



DISTRIBUCIÓN F de Snedecor ($F_{n_1, n_2}^{(p)}$)

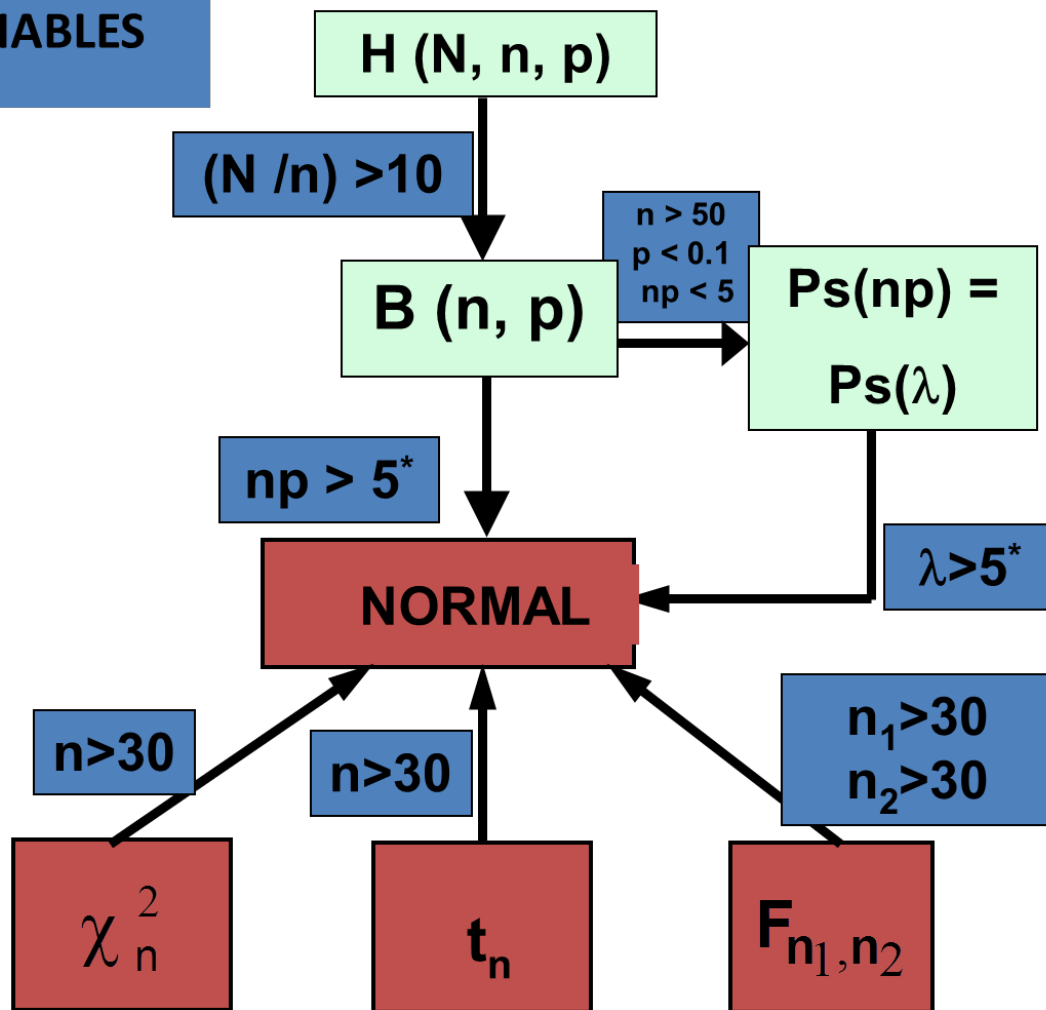


	Grados de libertad de la varianza mayor															
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
p→	0.05	0.01	0.05	0.01	0.05	0.01	0.05	0.01	0.05	0.01	0.05	0.01	0.05	0.01	0.05	0.01
1	161.45	4052.2	199.50	4999.3	21.71	5403.5	224.58	5624.3	230.16	5763.9	233.99	5858.9	236.77	5928.3	238.88	5980.9
2	18.51	98.50	19.00	99.00	19.16	99.16	19.25	99.25	19.30	99.30	19.33	99.33	19.35	99.36	19.37	99.38
3	10.13	34.12	9.55	30.82	9.28	29.46	9.12	28.71	9.01	28.24	8.94	27.91	8.89	27.67	8.85	27.49
4	7.71	21.20	6.94	18.00	6.59	16.69	6.39	15.98	6.26	15.52	6.16	15.21	6.09	14.98	6.04	14.80
5	6.61	16.26	5.79	13.27	5.41	12.06	5.19	11.39	5.05	10.97	4.95	10.67	4.88	10.46	4.82	10.29
6	5.99	13.75	5.14	10.92	4.76	9.78	4.53	9.15	4.39	8.75	4.28	8.47	4.21	8.26	4.15	8.10
7	5.59	12.25	4.74	9.55	4.35	8.45	4.12	7.85	3.97	7.46	3.87	7.19	3.79	6.99	3.73	6.84
8	5.32	11.26	4.46	8.65	4.07	7.59	3.84	7.01	3.69	6.63	3.58	6.37	3.50	6.18	3.44	6.03
9	5.12	10.56	4.26	8.02	3.86	6.99	3.63	6.42	3.48	6.06	3.37	5.80	3.29	5.61	3.23	5.47
10	4.95	10.04	4.10	7.59	3.71	6.55	3.48	5.99	3.33	5.64	3.22	5.39	3.14	5.20	3.07	5.06
11	4.84	9.65	3.98	7.21	3.59	6.22	3.36	5.67	3.20	5.32	3.09	5.07	3.01	4.89	2.95	4.74
12	4.75	9.33	3.89	6.93	3.49	5.95	3.26	5.41	3.11	5.06	3.00	4.82	2.91	4.64	2.85	4.50
13	4.67	9.07	3.81	6.70	3.41	5.74	3.18	5.21	3.03	4.86	2.92	4.62	2.83	4.44	2.77	4.30
14	4.60	8.86	3.74	6.51	3.34	5.56	3.11	5.04	2.96	4.69	2.85	4.46	2.76	4.28	2.70	4.14
15	4.54	8.68	3.68	6.36	3.29	5.42	3.06	4.89	2.90	4.56	2.79	4.32	2.71	4.14	2.64	4.00
16	4.49	8.53	3.63	6.23	3.24	5.29	3.01	4.77	2.85	4.44	2.74	4.20	2.66	4.03	2.59	3.89
17	4.45	8.40	3.59	6.11	3.20	5.16	2.96	4.67	2.81	4.34	2.70	4.10	2.61	3.93	2.55	3.76

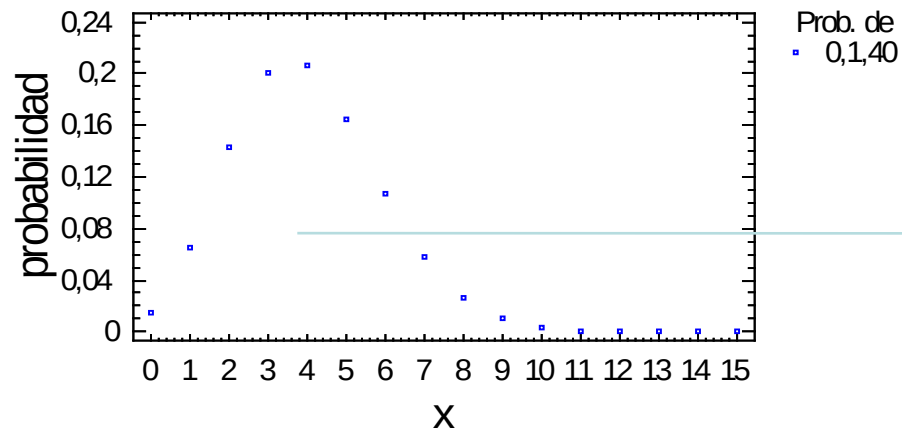
$$X \equiv F_{3,10}$$

$$F_{3,10(0.05)} = 3.71$$

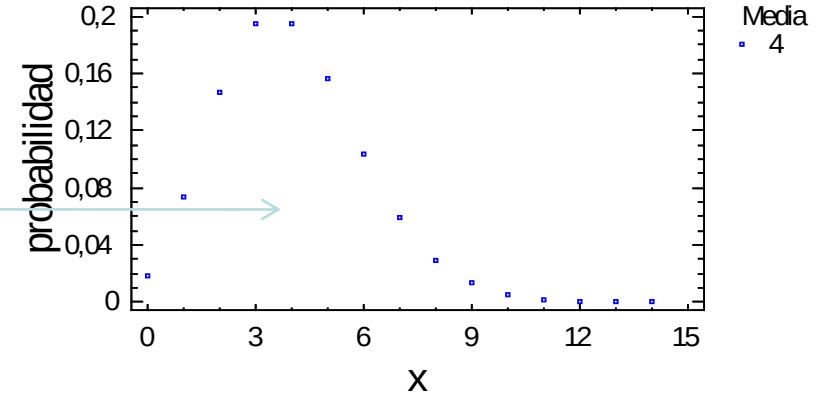
APROXIMACIONES ENTRE VARIABLES



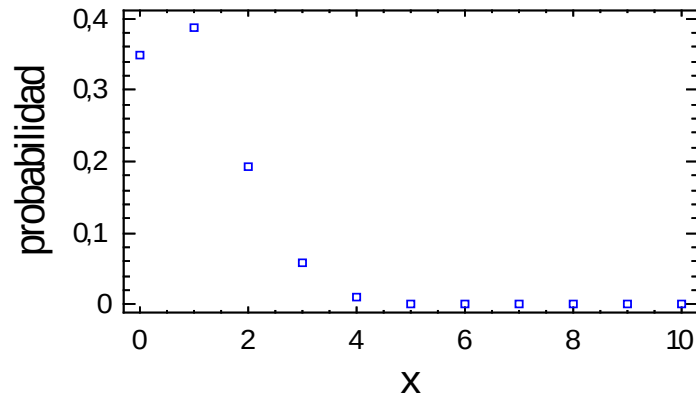
Binomial Distribución



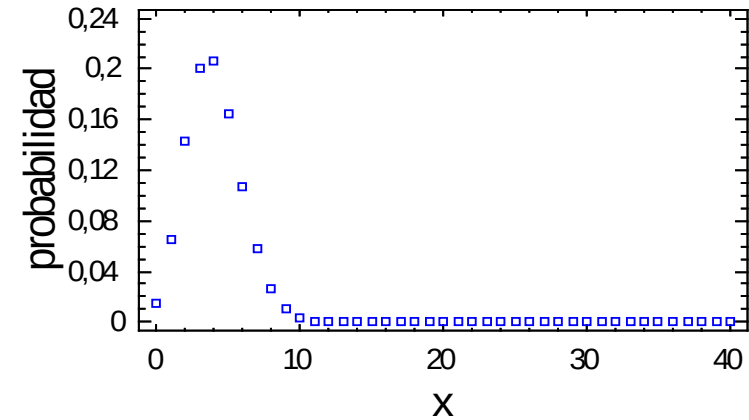
Poisson Distribución



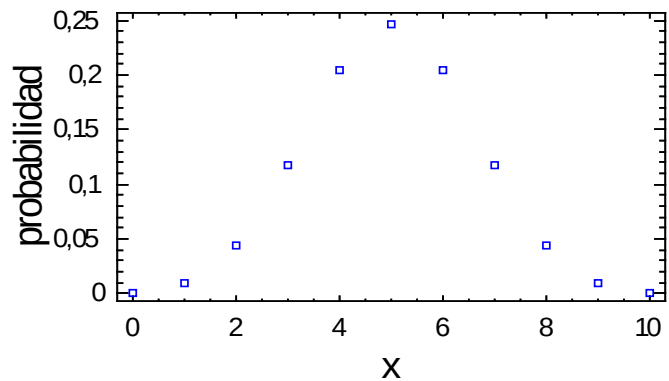
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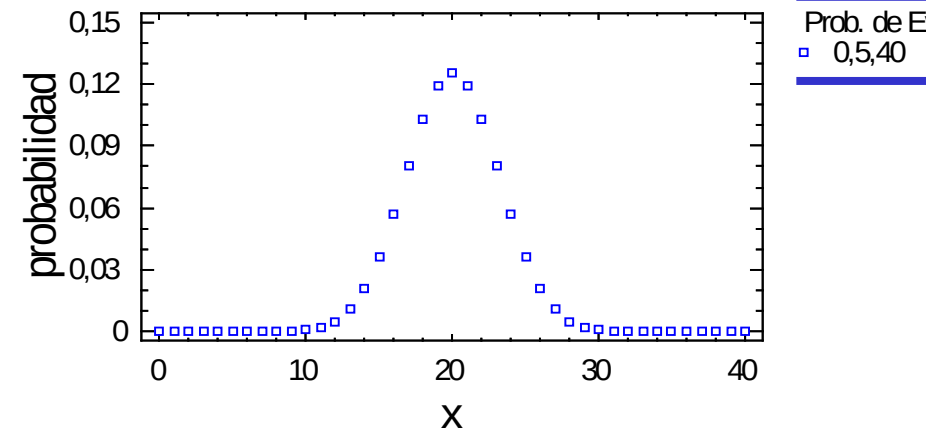
Binomial Distribución



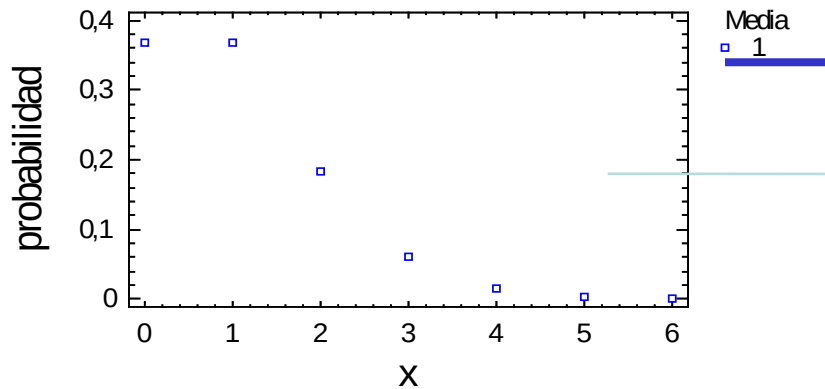
Binomial Distribución



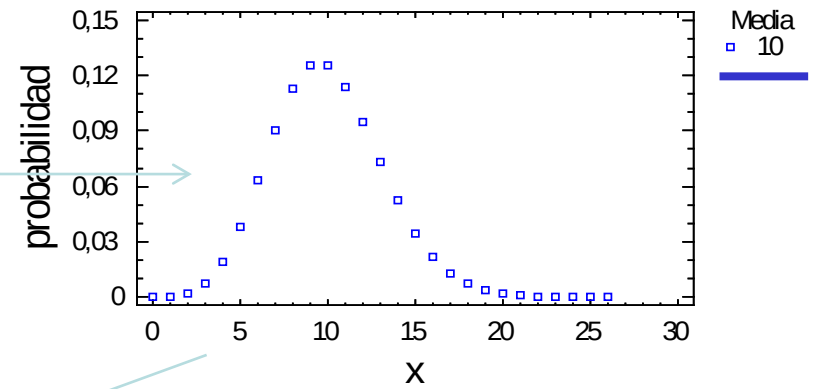
Binomial Distribución



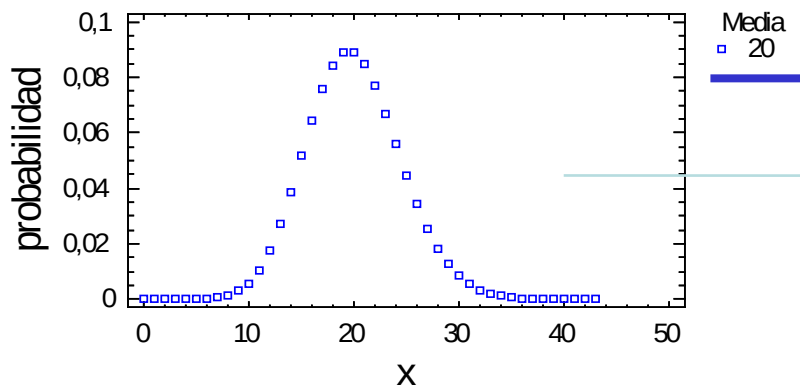
Poisson Distribución



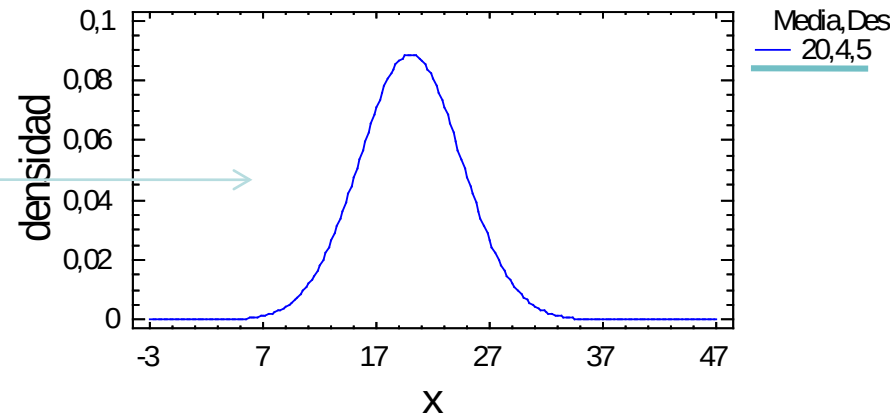
Poisson Distribución



Poisson Distribución



Normal Distribución

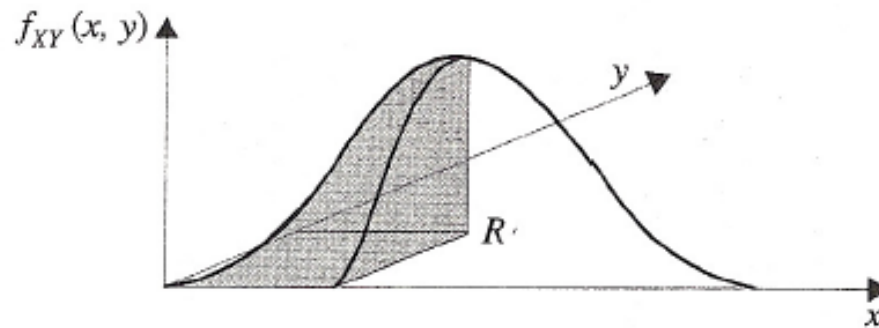


- It is the two-dimensional variable (Y_1, Y_2) whose density function is

$$f_y(\vec{y}) = \frac{1}{2\pi \|V_y\|^{1/2}} e^{-\frac{1}{2}(\vec{y}-\vec{\mu}_Y)' V_y^{-1}(\vec{y}-\vec{\mu}_Y)}$$

$$f_y(\vec{y}) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{(1-\rho^2)}} e^{-\frac{1}{2}(\vec{y}-\vec{\mu}_Y)' V_y^{-1}(\vec{y}-\vec{\mu}_Y)}$$

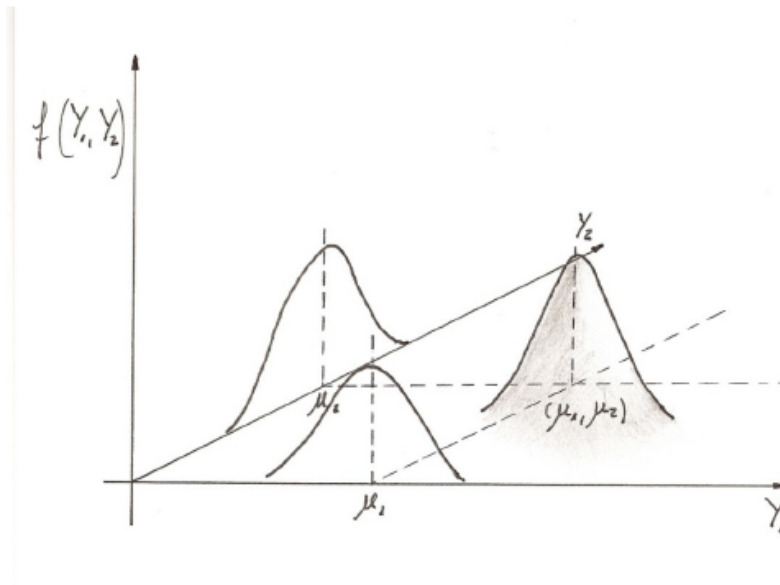
$$\text{Con} \quad V_y = \begin{bmatrix} \sigma_1^2 & \sigma_{1,2}^2 \\ \sigma_{2,1}^2 & \sigma_2^2 \end{bmatrix} \quad \vec{\mu}_Y = \begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}$$



The probability of that (X, Y) is in the region R is determined by the volume of $f_{xy}(x,y)$ in the region R

- Of a two-dimensional normal $N(Y_1, Y_2)$, both marginal are one-dimensional normal with averages and variances respectively

$$Y_1 = N(\mu_1, \sigma_1), \quad Y_2 = N(\mu_2, \sigma_2)$$



Consider the variable (height, weight). If we look now the weight of all those who have a certain stature (i.e. 170), this weight is not constant but varies according to a certain pattern (is a one-dimensional random variable) that will have an average and a certain variance. For other (e.g. 185) height value, the pattern of variation will be possibly different with probably greater mean.

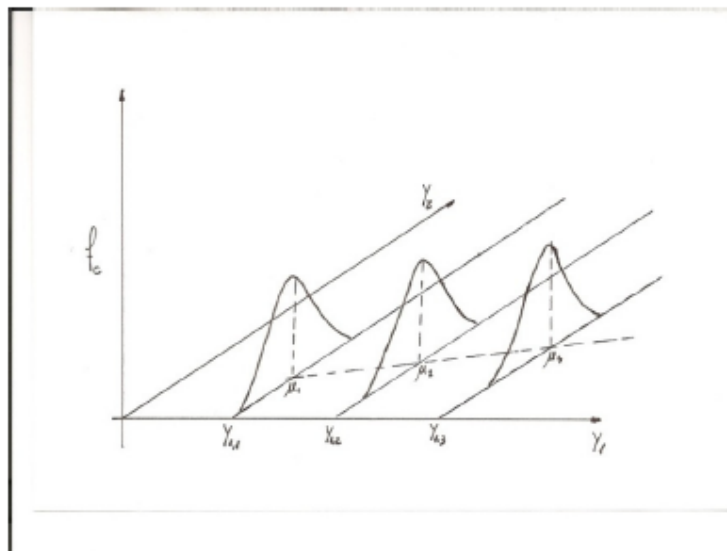
The one-dimensional distribution that has an of them components of a variable two-dimensional, when the other takes a value determined, (**or takes values in a differential interval determined**), is called conditional distribution Y given x,

(Y/x):

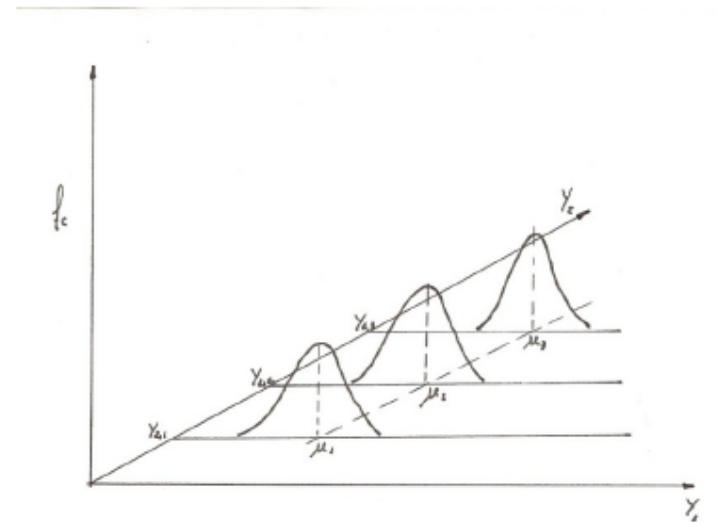
- Conditional distribution of Y given x
- Distribution of Y conditioned by x
- Distribution of Y when X vale x

- In the case of two-dimensional normal, conditional distributions are also normal

$$y_2/y_1$$



$$y_1/y_2$$



Parameters

$$\underline{E(Y_2/y_1)} = m_2 + \frac{\text{cov}(Y_1, Y_2)}{\sigma_1^2} (y_1 - m_1)$$

The conditional average depends on the value of the conditioning variable (except independence between variables)

$$\underline{D^2(Y_2/y_1)} = \sigma_2^2 \cdot (1 - \rho^2)$$

???

The variance of the conditional not depends of the value of the conditioning variable. All variances are equal. This property are called homoscedasticity

- The proportion of cement, X , that it is added to the concrete and the drying time, Y (hours), it is known that follows a two-dimensional Normal distribution of vector of means and matrix of variances,

$$\bar{\mu} = \begin{Bmatrix} 0,3 \\ 9 \end{Bmatrix} \quad \mathbf{V} = \begin{bmatrix} 0,05 & 0,25 \\ 0,25 & 2 \end{bmatrix}$$

- if the proportion added of cement is of 0.25, what is the probability of that the drying time lasts more than 10 hours?
- If the drying time is 10 hours, what is the probability that the ratio of cement is less than 0.6?
- What proportion of cement we have to add so drying time is less than 10 hours in 90% of cases?